

Graduation Project Proposal

Project Title: FastAF

Project Members:

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Introduction: Pharmacies often rely on a single cashier to handle all transactions, which creates a bottleneck and increases the risk of delays and long queues or disruptions. Our proposed system aims to eliminate this single point of failure by decentralizing the cashiering process. Instead of relying on one cashier, our system allows pharmacists working in the pharmacy to handle sales directly from their mobile devices and without the need of the barcode. This will significantly improve efficiency and streamline the checkout process.

Problem Statement:

Pharmacies commonly face issues with slow transactions due to dependence on a single cashier system. This results in customer dissatisfaction and operational inefficiencies. By enabling multiple authorized personnel to process sales simultaneously through their mobile devices, we aim to eliminate this bottleneck and enhance service delivery.

Objectives:

- Develop a web-based platform for pharmacy management, including pharmacy registration, employee account creation, inventory management, and sales analytics.
- Develop an Android application to enable doctors working in the pharmacy to handle cashiering independently using their mobile devices.
- Implement a real-time database for seamless stock updates and sales tracking.
- Enable camera-based scanning for quick identification and addition of medical items to the cart without the need of barcode.
- Integrate a receipt printing system that updates the pharmacy's database upon checkout.
- Provide data visualization dashboards to aid in decision-making.
- Develop a data collection application to gather labeled images of drug items for training the recognition system.

System Features:

Data Collection Application:

- Used to collect labeled images of medical items to train the image recognition model.
- Helps improve accuracy in scanning and identifying drug items.

Website:

1. **Home Page:** Introduction to the system and sign-up/log-in options.
2. **Pharmacy Profile Management:** Owners can create and manage pharmacy accounts.
3. **Employee Management:** Pharmacy owners can create accounts for employees.
4. **Inventory Management:** Maintain drug stock with details such as name, quantity, and pricing.
5. **Sales Analytics and Reports:** Visual dashboards for tracking pharmacy sales and performance metrics.

Android Application:

1. **User Authentication:** Employees log in to access the system.
2. **Item Scanning:** Utilize the phone camera to scan medical items.
3. **Interactive Selection:** Highlight recognized items and allow easy selection (tap a recognized item to add to the cart or long touch for information about that item).
4. **Cart Management:** Pharmacists can add items to a cart, adjust quantities, and proceed to checkout.
5. **Checkout & Receipt Generation:** Update the database and send a receipt request to the pharmacy's printer.
6. **Stock Updating:** Pharmacists can add more stock when necessary.

Technologies Used:

- **Frontend:**
 - Language : JavaScript
 - Framework : React, axios
- **Backend:**
 - language : java

- framework : Spring Boot
- DataBase : PostgreSQL MongoDB Redis Elastic
- **UI\UX:**
 - Framework : figma
- **Application:**
 - Language : Java/kotlin for Android developing
 - Tools : Android studio code
- **Machine Learning:**
 - language : python
 - framework : tensorflow/keras/scikit-learn, yolo, OpenCV
- **Version control:** git and GitHub
- **Hardware Integration:** Thermal printers for receipt printing.

Expected Outcomes:

- Improved efficiency in pharmacy cashiering by decentralizing the process.
- Reduced wait times for customers due to multiple checkout points.
- Enhanced inventory management and sales tracking.
- A functional prototype with a user-friendly interface.
- Dispense with the barcode and identify the item through the item itself

Project phases:

1. **Phase 1 - Research & Planning:** understanding the problem and technology selection.
2. **Phase 2 - Pre data Collection:** Develop and deploy the data collection app.
3. **Phase 3 - Data collection:** collect the data using the data collection app
4. **Phase 4 – model training:** train a machine learning model with the collected data
5. **Phase 5 - Development:** Implement website, Android app, and backend services.
6. **Phase 6 - Testing & Refinement:** Conduct testing, debugging, and refinements.
7. **Phase 7 - Deployment & Documentation:** Deploy system and finalize project documentation.